

Ejemplo de configuración real: B210_AIR_ue_rf.conf

```
#Los valores de gain para la gNb en este setup fueron de Tx=70, Rx=30
[rf]
freq_offset = -2000
tx_gain = 70
rx_gain = 30
srate = 23.04e6
nof_antennas = 1

device_name = uhd
device_args = sync=internal,type=b200,num_recv_frames=64,num_send_frames=64
time_adv_nsamples = auto

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0
#dl_freq = 1842.5e6
#ul_freq = 1747.5e6

[rat.nr]
bands = 3
nof_carriers = 1
max_nof_prb = 106
nof_prb = 106
#dl_nr_arfcn = 173800
#ssb_nr_arfcn = 172330

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = e734f8734007d6c5ce7a0508809e7e9c
k = 8baf473f2f8fd09487cccbd7097c6862
imsi = 001010000000003
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = true
```

Ejemplo de configuración real: B210_AIR_ue_rf_n28.conf

```
#Los valores de gain para la gNb en este setup fueron de Tx=70, Rx=30
#11/3/2025 quedó en 5Mhz de bw
[rf]
freq_offset = 500
tx_gain = 70
rx_gain = 40
srate = 23.04e6
nof_antennas = 1
device_name = uhd
device_args = sync=internal
#time_adv_nsamples = 40

[rat.eutra]
dl_earfcn = 2850
nof_carriers = 0
#dl_freq = 1842.5e6
#ul_freq = 1747.5e6

[rat.nr]
bands = 28
nof_carriers = 1
max_nof_prb = 25
nof_prb = 25
dl_nr_arfcn = 154000
ssb_nr_arfcn = 154090

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 504f20634f6320504f50206363500a4f
k = 6874736969202073796d4b2079650a73
imsi = 0010100000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = true
```

Ejemplo de configuración real: ue.conf

```
[rf]
freq_offset = 0
tx_gain = 80

device_name = zmq
device_args = tx_port=tcp://*:2001,rx_port=tcp://localhost:2000,id=ue,base_srate=23.04e6

#####
[rat.eutra]
dl_eearfcn = 3350
#nof_carriers = 1

#####
[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

#####
[log]
all_level = warning
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

#####
[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 001010123456780
imei = 353490069873319
#reader =
#pin = 1234

[rrc]
#ue_category = 4
#release = 8
#feature_group = 0xe6041000
#mbms_service_id = -1
#mbms_service_port = 4321

[nas]
#apn = internetinternet
#apn_protocol = ipv4
#user = srsuser
#pass = srspass
#force_imsi_attach = false
#eia = 1,2
#eea = 0,1,2

[gw]
#netns =
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0
```

```
[gui]
enable = false

[channel.dl]
#enable = false

[channel.dl.awgn]
#enable = false
#snr = 30

[channel.dl.fading]
#enable = false
#model = none

[channel.dl.delay]
#enable = false
#period_s = 3600
#init_time_s = 0
#maximum_us = 100
#minimum_us = 10

[channel.dl.rlf]
#enable = false
#t_on_ms = 10000
#t_off_ms = 2000

[channel.dl.hst]
#enable = false
#period_s = 7.2
#fd_hz = 750.0
#init_time_s = 0.0

[channel.ul]
#enable = false

[channel.ul.awgn]
#enable = false
#n0 = -30

[channel.ul.fading]
#enable = false
#model = none

[channel.ul.delay]
#enable = false
#period_s = 3600
#init_time_s = 0
#maximum_us = 100
#minimum_us = 10

[channel.ul.rlf]
#enable = false
#t_on_ms = 10000
#t_off_ms = 2000

[channel.ul.hst]
#enable = false
#period_s = 7.2
#fd_hz = -750.0
#init_time_s = 0.0

[phy]
#rx_gain_offset = 62
#prach_gain = 30
#cqi_max = 15
#cqi_fixed = 10
#snr_ema_coeff = 0.1
```

```
#snr_estim_alg      = refs
#pdsch_max_its      = 8      # These are half iterations
#pdsch_meas_evm      = false
#nof_phy_threads     = 3
#equalizer_mode      = mmse
#correct_sync_error  = false
#sfo_ema             = 0.1
#sfo_correct_period  = 10
#sss_algorithm       = full
#estimator_fil_auto  = false
#estimator_fil_stddev = 1.0
#estimator_fil_order = 4
#snr_to_cqi_offset   = 0.0
#interpolate_subframe_enabled = false
#pdsch_csi_enabled   = true
#pdsch_8bit_decoder  = false
#force_ul_amplitude  = 0
#detect_cp           = false
```

```
#in_sync_rsrp_dbm_th = -130.0
#in_sync_snr_db_th    = 3.0
#nof_in_sync_events   = 10
#nof_out_of_sync_events = 20
```

```
#force_N_id_2        = 1
#force_N_id_1        = 10
```

```
[phy.nr]
#store_pdsch_ko = false
```

```
[cfr]
#enable          = false
#mode            = manual
#manual_thres     = 2.0
#strength        = 1.0
#auto_target_papr = 7.0
#ema_alpha       = 0.0143
```

```
[sim]
#airplane_t_on_ms = -1
#airplane_t_off_ms = -1
```

```
[general]
#metrics_csv_enable = false
#metrics_period_secs = 1
#metrics_csv_filename = /tmp/ue_metrics.csv
#have_tti_time_stats = true
#tracing_enable      = true
#tracing_filename    = /tmp/ue_tracing.log
#tracing_buffcapacity = 1000000
#metrics_json_enable = false
#metrics_json_filename = /tmp/ue_metrics.json
```

Ejemplo de configuración real: ue_887_srate_2304_20Mhz.conf

```
[rf]
freq_offset = 0
tx_gain = 10
rx_gain = 5
srate = 23.04e6
nof_antennas = 1

device_name = uhd
device_args = sync=internal
time_adv_nsamples = auto

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0
#dl_freq = 1842.5e6
#ul_freq = 1747.5e6

[rat.nr]
bands = 5
nof_carriers = 1
max_nof_prb = 106
nof_prb = 106
dl_nr_arfcn = 177400
ssb_nr_arfcn = 177370

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 001010123456780
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = true

#[expert]
#lte_sample_rates = true
```


Ejemplo de configuración real: ue_935_srate_2304_20Mhz.conf

```
[rf]
freq_offset = 0
tx_gain = 10
rx_gain = 5
srate = 23.04e6
nof_antennas = 1

device_name = uhd
device_args = sync=internal
time_adv_nsamples = auto

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0
#dl_freq = 1842.5e6
#ul_freq = 1747.5e6

[rat.nr]
bands = 8
nof_carriers = 1
max_nof_prb = 106
nof_prb = 106
dl_nr_arfcn = 187000
ssb_nr_arfcn = 186970

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 001010123456780
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = true

#[expert]
#lte_sample_rates = true
```


Ejemplo de configuración real: ue_1842_5_srate_5,76_5Mhz.conf

```
[rf]
freq_offset = 0
tx_gain = 10
rx_gain = 10
srate = 5.76e6
nof_antennas = 1

device_name = uhd
device_args = sync=internal
time_adv_nsamples = auto

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0
#dl_freq = 1842.5e6
#ul_freq = 1747.5e6

[rat.nr]
bands = 3
nof_carriers = 1
max_nof_prb = 25
nof_prb = 25
#dl_nr_arfcn = 368500
#ssb_nr_arfcn = 367930

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 001010123456780
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = true

#[expert]
#lte_sample_rates = true
```


Ejemplo de configuración real: ue_1842_5_srate_1152_10Mhz.conf

```
[rf]
freq_offset = 0
tx_gain = 10
rx_gain = 5
srate = 11.52e6
nof_antennas = 1

device_name = uhd
device_args = sync=internal
time_adv_nsamples = auto

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0
#dl_freq = 1842.5e6
#ul_freq = 1747.5e6

[rat.nr]
bands = 3
nof_carriers = 1
max_nof_prb = 52
nof_prb = 52
#dl_nr_arfcn = 368500
#ssb_nr_arfcn = 367930

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 001010123456780
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = true

#[expert]
#lte_sample_rates = true
```


Ejemplo de configuración real: X310_AIR_ue_rf.conf

```
[rf]
freq_offset = 0
tx_gain = 20
rx_gain = 10
srate = 23.04e6
nof_antennas = 1

device_name = uhd
device_args = sync=internal
time_adv_nsamples = auto

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0
#dl_freq = 1842.5e6
#ul_freq = 1747.5e6

[rat.nr]
bands = 3
nof_carriers = 1
max_nof_prb = 106
nof_prb = 106
#dl_nr_arfcn = 368500
#ssb_nr_arfcn = 367930

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = debug
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 001010123456780
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = true
```

Ejemplo de configuración real: X310_AIR_ue_rf_n28.conf

```
#Los valores de gain para la gNb en este setup fueron de Tx=70, Rx=30
[rf]
freq_offset = -3000
tx_gain = 20
rx_gain = 10
srate = 23.04e6
nof_antennas = 1

device_name = uhd
device_args = sync=internal
time_adv_nsamples = auto

[rat.eutra]
dl_earfcn = 2850
nof_carriers = 0
#dl_freq = 1842.5e6
#ul_freq = 1747.5e6

[rat.nr]
bands = 28
nof_carriers = 1
max_nof_prb = 106
nof_prb = 106
dl_nr_arfcn = 154000
ssb_nr_arfcn = 152650

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc  = e734f8734007d6c5ce7a0508809e7e9c
k    = 8baf473f2f8fd09487cccbd7097c6862
imsi = 001010000000003
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = true
```